National Aeronautics and
Space Administration

Synthetic transcriptomics for mouse spaceflight

Can generated RNA-seq improve FLT vs GC analysis within each tissue?

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QUESTION

Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

NASA OSDR

1,610 biological profiles

75 accessions

835 FLT

775 GC

ARCHS4 mouse

997,515 profiles audited

17,244 selected

20 tissues

- Mission, strain, material and assay differences can resemble a flight effect.
- The generator must preserve tissue and FLT/GC structure without memorizing samples.
- Statistical evidence must still come from observed OSDR profiles.

Sources: NASA OSDR Biological Data API; ARCHS4 mouse v2.5.

METHOD

We built a configurable bulk RNA-seq generation pipeline

Data scope, transforms, harmonization, training and conditioning can change without changing the evaluation contract.

Data + scope

OSDR API or ARCHS4
one or many studies
pooled or per tissue

Transform

raw / CPM / TPM
log + scaling
all genes / HVG / L1000

Harmonize

none or study z
ComBat / MBatch
MOBER adapter

Model + train

WGAN-GP or DDIM
OSDR or ARCHS4
pretrain + adapt

Condition

FLT/GC + tissue
study + material
available covariates

Selected branch

TPM / MaxAbs -> ARCHS4 DDIM -> OSDR adaptation. Conditions: tissue, FLT/GC, accession and material; no global batch correction.

A 463-row experiment plan and nine matched liver harmonization arms were screened before full training.

Generator sources: Vinas et al., Bioinformatics (2022); Lacan et al., BMC Bioinformatics (2026).

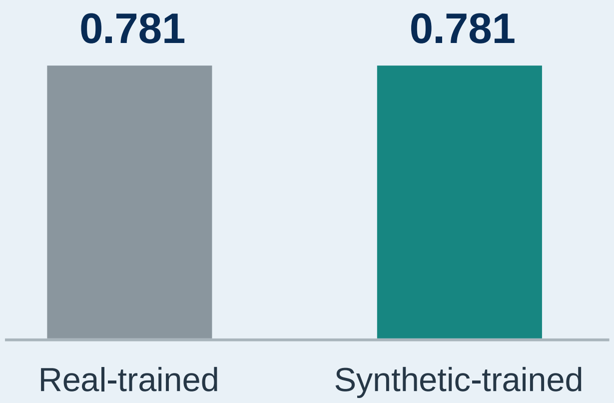
VALIDATION

DDIM matched expression and reduced separability

AA closer to 0.5 means a classifier has less success separating real and generated profiles; lower FD is better.

ARCHS4 tissue probe

Balanced accuracy on held-out real profiles



Metric	WGAN-GP	DDIM	Reading
Correlation	0.976	0.974	similar
F1	0.985	0.997	higher
Adversarial accuracy	0.636	0.475	closer to 0.5
FD / real P95	0.144	0.074	lower

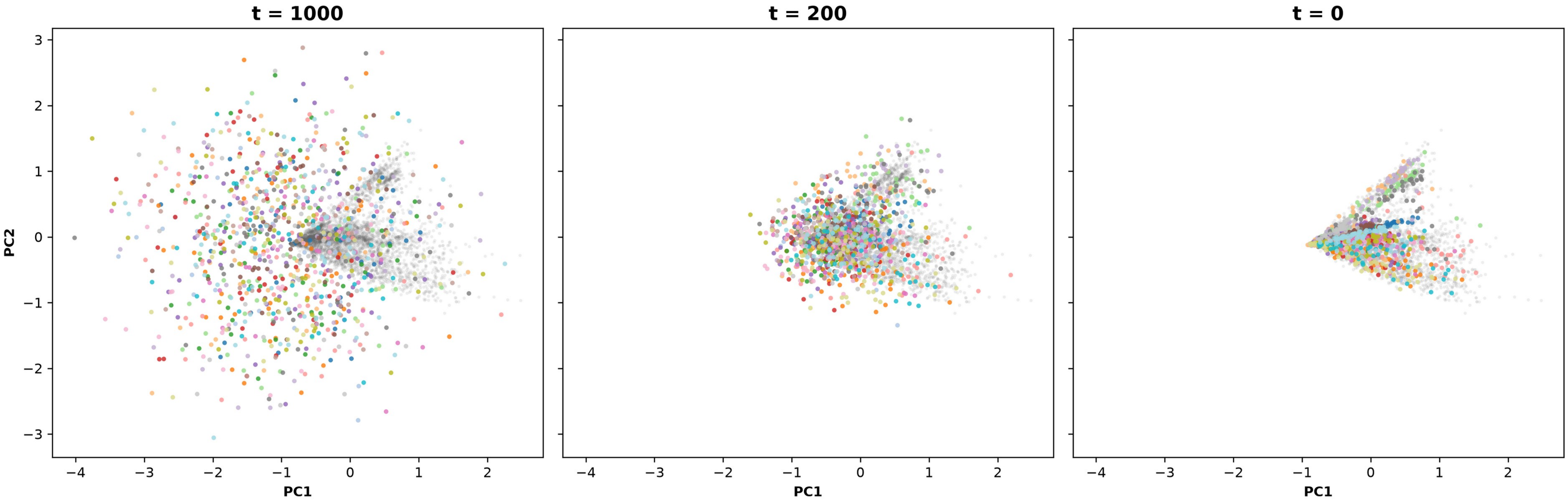
Use DDIM downstream

Lower separability and distributional distance, with comparable correlation and higher F1.

WGAN values use validation data and DDIM values use the stated OSDR test; they are not paired on one common split.

Diffusion learns tissue structure from noise

The same PCA axes follow 1,024 generated profiles through reverse diffusion.



t = 1000: noise

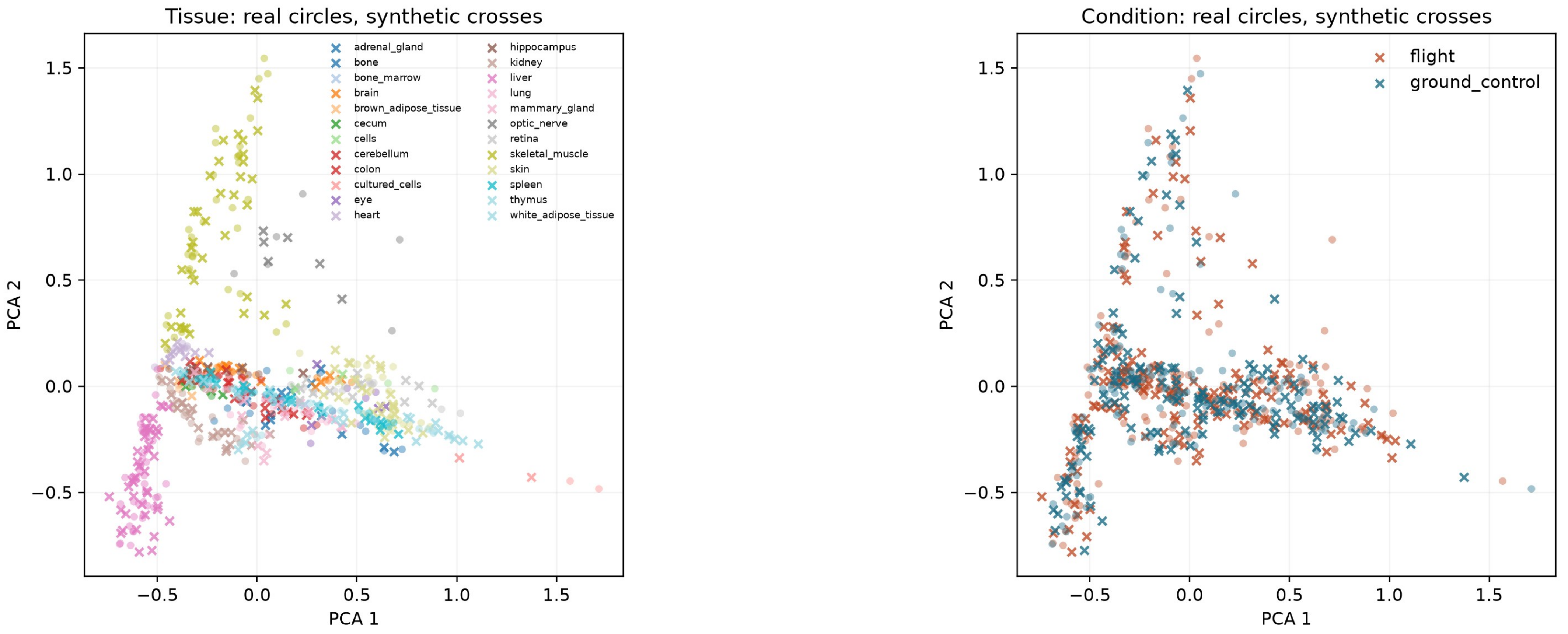
t = 200: partial structure

t = 0: tissue-conditioned profiles

Gray points are real ARCHS4 profiles; colors identify generated tissue conditions. PC1 and PC2 limits are shared across panels.

Generated profiles track the real OSDR PCA manifold

Circles are locked real OSDR profiles; crosses are matched DDIM profiles from generation seed 5020.



Interpretation: tissue structure dominates; FLT and GC are subtler. PCA is descriptive, so quantitative metrics determine validation.

PCA was fitted in the common locked OSDR expression space. This is PCA, not UMAP.

Synthetic profiles entered the analysis in five different ways

Every arm was judged on held-out real profiles before synthetic attribution was allowed.

Real OSDR
observed FLT / GC

DDIM generated
matched conditions

Real only
rank: real
fit: real

Generated only
rank: generated
fit: generated

Real + generated
rank: consensus
fit: equal weight

Guided, real fit
rank: consensus
fit: real

Guided, 5% synth.
rank: consensus
fit: real + 5%

Held-out real profiles

BA | AUROC | AP

select one eligible arm
for each tissue

Biology check FLT vs GC effects, random-effects meta-analysis and BH FDR use real OSDR profiles only.

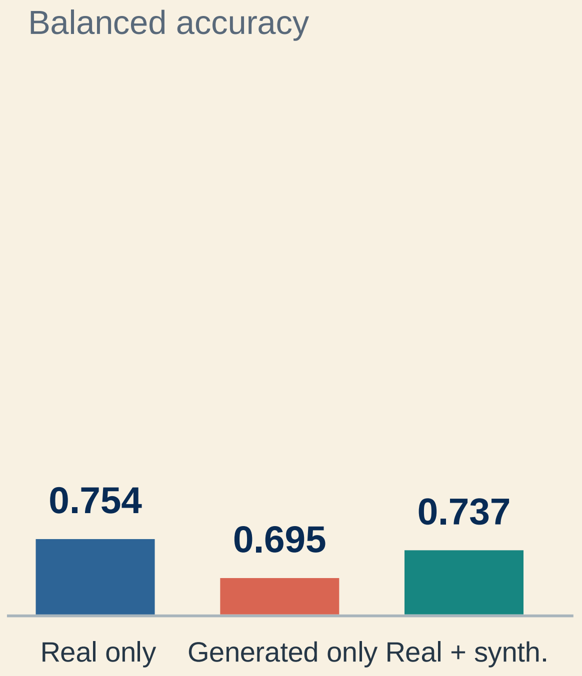
Synthetic profiles never increase animal n.

UTILITY

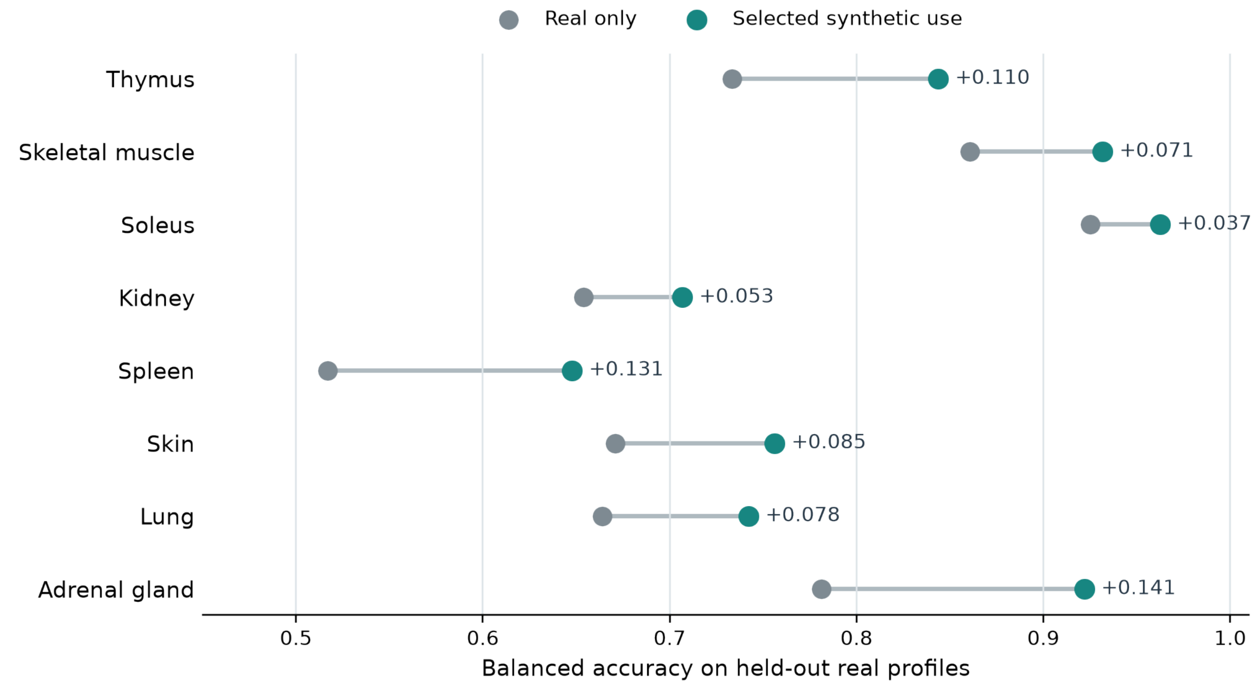
Pooling tissues hid useful signal

A single FLT/GC classifier did not improve, but tissue-specific synthetic use often did.

Pooled across tissues



Selected use within each tissue



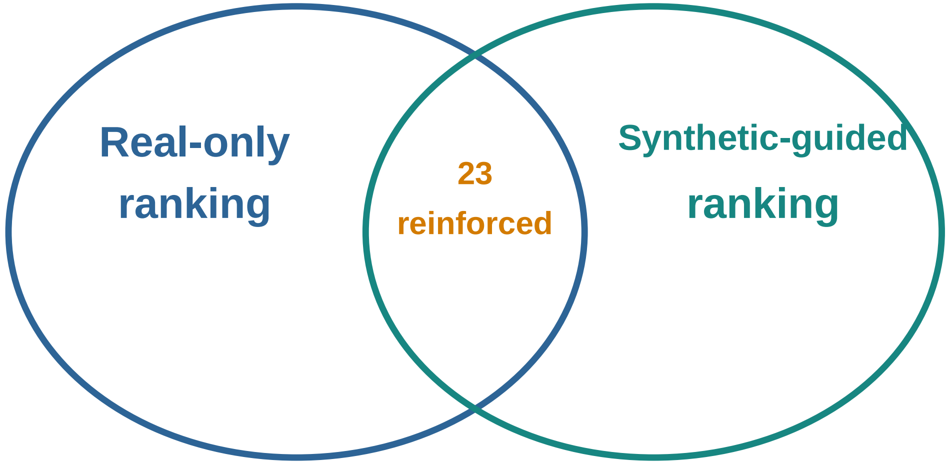
The useful arm differed by tissue.

Development results with held-out profiles from represented accessions; they do not measure transfer to a new mission.

Synthetic guidance changed ranking, not statistical evidence

Promoted and reinforced describe repeated feature selection; both require a supporting effect in real OSDR data.

Repeated stable selection



26 promoted
selected only with synthetic guidance

Real OSDR association test

- Estimate FLT vs GC inside each accession
- Combine accession effects with a random-effects model
- Apply BH FDR within each tissue

49 synthetic-informed associations

These are 49 tissue-gene associations among 459 BH-FDR rows. A gene can appear in more than one tissue.

LITERATURE INTERPRETATION

Annotation separates recovery from hypothesis extension

The label describes the relationship to prior evidence, not whether a gene is biologically plausible.

1 Candidate
Gene + tissue + FLT direction

2 Search
Spaceflight + mechanism literature

3 Classify
Relation to published evidence

4 Interpret
Recovery, extension, or unmatched

What the 26 promoted associations showed

11	Aligning	3 exact matches + 8 same-tissue process matches
13	Complementary	Related process or mechanism; not direct replication
1	Ambiguous	Birc5 differed across mission contexts
1	Literature-unmatched	Psmb8 remains mechanistically plausible

How to read a result

- Promoted: synthetic guidance changed stable ranking
- BH FDR: the association is present in real OSDR data
- Literature class: direct, related, mixed, or unmatched
- Interpretation: a testable hypothesis, not independent proof

11 align and 13 extend prior biology; only 3 are exact gene-tissue-direction matches.

Targeted review completed 2026-08-03; full decisions and 19-source inventory are in Supplementary Tables S16-S17.

GENE INVENTORY

Synthetic-informed genes spanned 10 tissue analyses

All 49 associations passed BH FDR in real OSDR data; synthetic profiles affected prioritization, not the statistical test.

P: promoted — aligning — complementary — ambiguous — literature-unmatched — **R: reinforced**

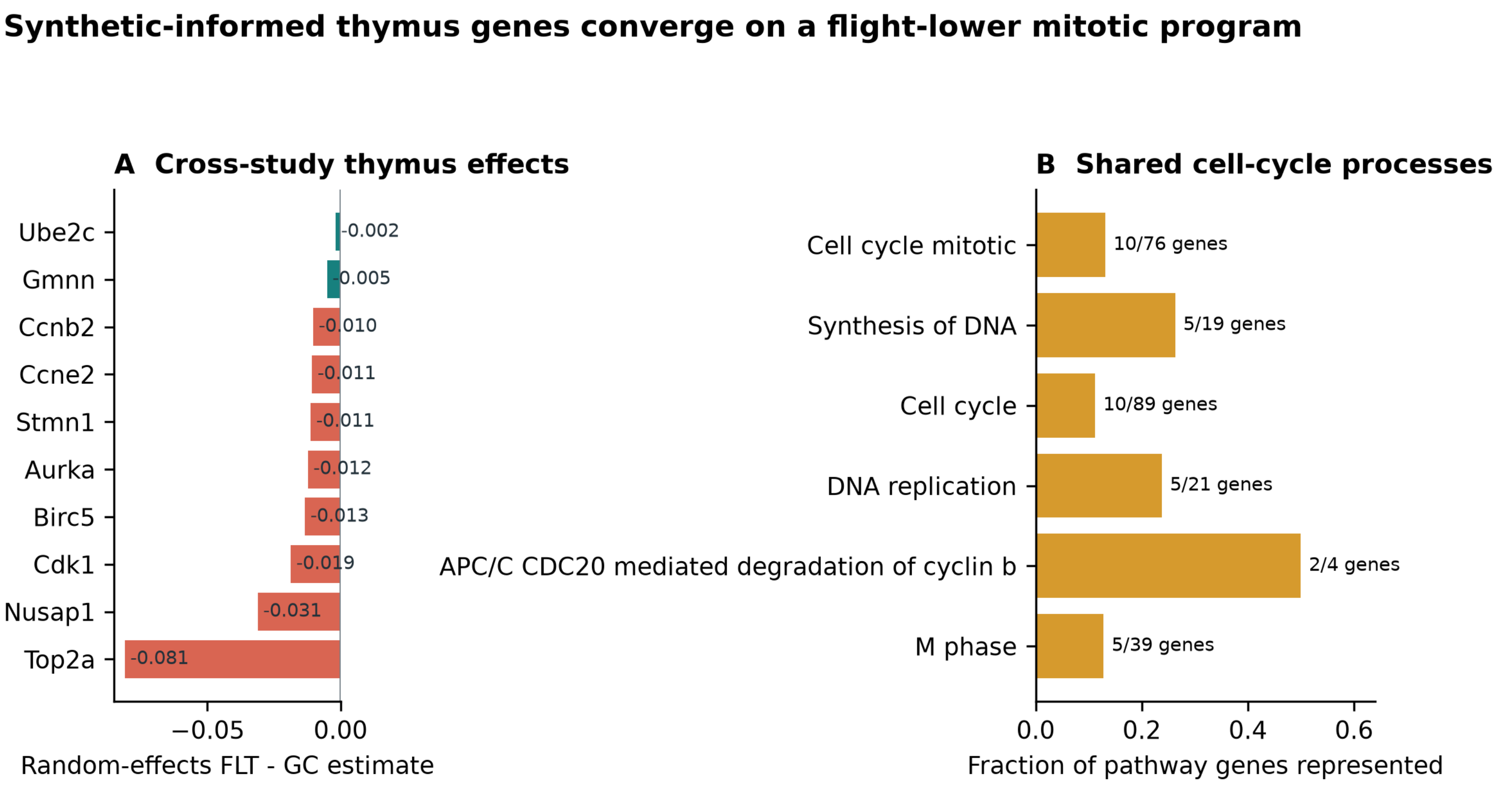
Direction is shown within each tissue.

Tissue	Direction	Genes
Thymus 16 genes	FLT higher	***P:Hsd17b11***, ***P:Etv1***, *R:Snx7*
	FLT lower	***P:Nusap1***, ***P:Stmn1***, ***P:Birc5***, ***P:Cdk1***, ***P:Top2a***, ***P:Ccnb2***, ***P:Aurka***, ***P:Ccne2***, ***P:Kif20a***, ***P:Pcna***, ***P:Ccnf***, *R:Ube2c*, *R:Gmnn*
Spleen 4 genes	FLT higher	***P:Rai14***, ***P:Myl9***, ***P:Ptprk***, *R:Loxl1*
	FLT lower	*none*
Kidney 2 genes	FLT higher	***P:Inpp4b***, *R:Slc37a4*
	FLT lower	*none*
Gastrocnemius 2 genes	FLT higher	***P:Nfkbia***
	FLT lower	***P:Fhl2***
Eye 1 gene	FLT higher	*none*
	FLT lower	*R:Klhl21*
Skeletal muscle (pooled) 12 genes	FLT higher	*R:Sox4*, *R:Cebpd*, *R:Sh3bp5*, *R:Prkcd*, *R:Arid5b*, *R:Sesn1*, *R:Tle1*
	FLT lower	***P:Klhl21***, ***P:Mapkapk5***, ***P:Reep5***, ***P:Itgb5***, *R:Bphl*
Soleus 5 genes	FLT higher	*R:Tpm1*
	FLT lower	*R:Bdh1*, *R:Ech1*, *R:Bnip3*, *R:Decr1*
Tibialis anterior 4 genes	FLT higher	***P:Cebpd***, *R:Cdkn1a*, *R:St3gal5*, *R:Bnip3*
	FLT lower	*none*
Adrenal gland 2 genes	FLT higher	*none*
	FLT lower	***P:Psmb8***, *R:Tspan4*
Skin 1 gene	FLT higher	***P:Plscr1***
	FLT lower	*none*

P/R reflects repeated feature selection. Literature colors apply to promoted genes only; full statistics are in Supplementary Table S10.

Thymus points to lower proliferative renewal

The core panel recovers cell-cycle biology; two flight-higher genes extend the hypothesis.



16 synthetic-informed BH-FDR associations

13 promoted | 3 reinforced

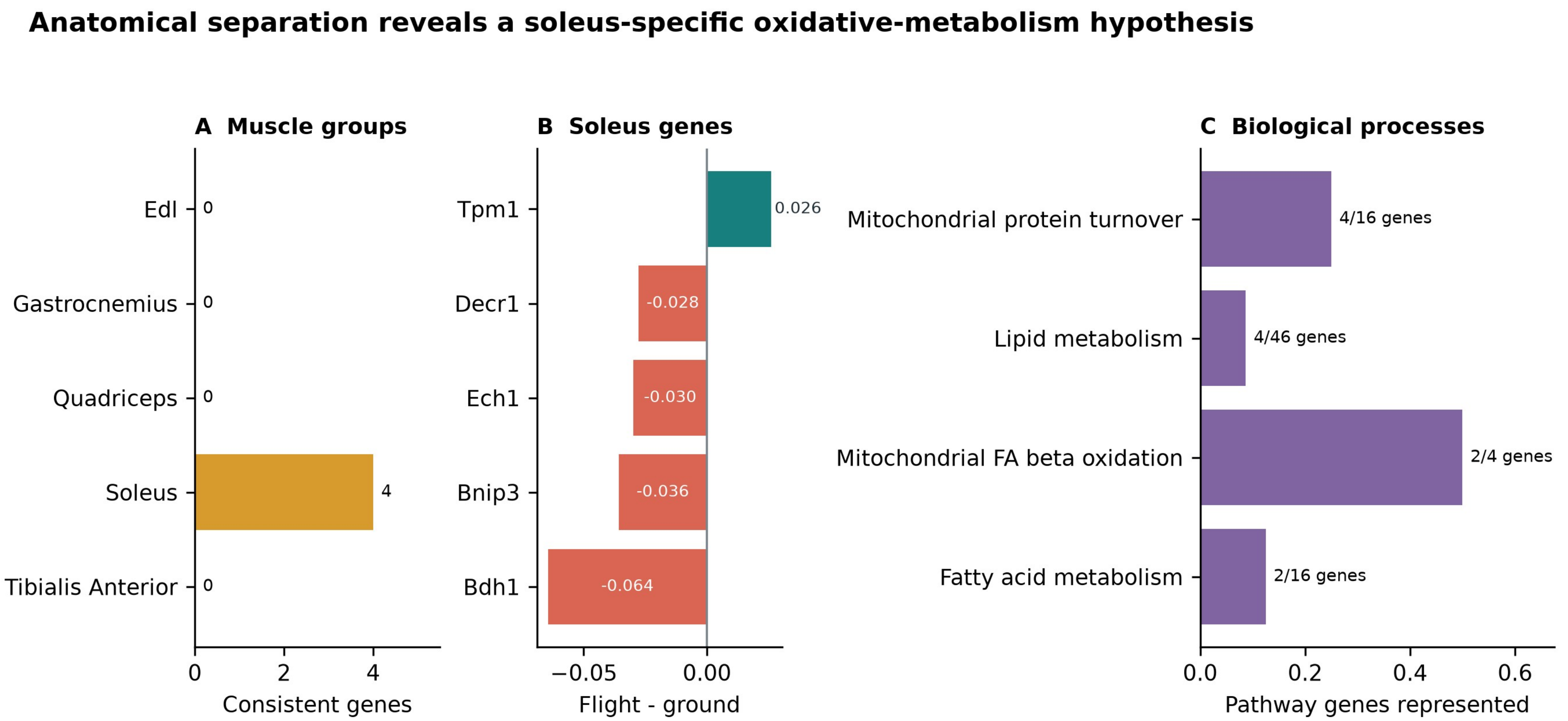
- Lower Nusap1, Stmn1, Birc5, Cdk1, Top2a and related genes
- Cell cycle, DNA replication, APC/C and G2/M processes
- Higher Hsd17b11 and Etv1 suggest lipid or T-cell-state shifts
- Bulk RNA-seq cannot separate cell loss from lower transcription

Context: Gridley et al. (2013), Horie et al. (2019), Keenan et al. (2025), and Shi et al. (2026).

SOLEUS

Soleus reinforces a mitochondrial and lipid program

This result strengthens an existing real-data pattern rather than promoting a new soleus gene.



0.925 -> 0.963

balanced accuracy

5 reinforced | 0 promoted

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Compatible with unloading and contractile remodeling

Literature context: Gambara et al. (2017) and Stein et al. (2002).

Kidney adds a focused signal; other tissues are exploratory

The strongest process-level stories were thymus and soleus, but several tissues produced narrower candidates.

Kidney	Inpp4b promoted, Slc37a4 reinforced; both higher in flight	**Focused secondary**
Spleen	Rai14, Ptprk and Myl9 promoted; Loxl1 reinforced; no Reactome enrichment	**Exploratory**
Skin + adrenal	Plscr1 higher in skin; adrenal Psmb8 is plausible but literature-unmatched	**Exploratory**
Lung + retina	Predictive gains without a synthetic-informed BH-FDR gene	**Prediction only**
Liver + EDL + quadriceps	Real-only arm retained	**No synthetic claim**

A plausible mechanism is not independent validation; these candidates still require targeted experiments.

TAKEAWAYS

Synthetic data helped most as a tissue-specific prior

The generator created useful structure, not new biological replication.

1 Generate

Conditional DDIM produced high-fidelity profiles with near-chance real-versus-synthetic separation.

2 Use

Pooled augmentation failed. Tissue-specific ranking and low-weight training were more useful.

3 Interpret

Thymus and soleus were clearest. Literature review separated recovery from complementary hypotheses.

Next test

Use independent samples and cell-resolved assays to test thymus proliferation, soleus metabolism, kidney signaling and adrenal Psmb8.

Generated profiles never enter the biological sample count or the BH-FDR test.

Thank you

Questions?

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Data	NASA OSDR | ARCHS4 | Reactome
Compute	NASA Ames Research Center
Code and manuscript	github.com/jasont314/nasa-mouse

All biological associations were tested in observed OSDR profiles.